HEALTH AI – SRS

Software Requirements Specification (SRS)

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the functional and non-functional requirements for the HEALTH AI Co-Creation & Innovation Platform. The document serves as the primary reference for the design, development, and testing phases of the project. It describes the system's intended behavior, constraints, user roles, use cases, and data model in detail. This SRS is prepared based on the HEALTH AI project brief provided for the SENG 384 course.

1.2 Scope

The HEALTH AI platform is a secure, GDPR-compliant web application that enables structured partner discovery between healthcare professionals and engineers in the health-tech innovation domain. The platform allows users to create structured announcements describing their collaboration needs, discover potential partners through search and filtering, initiate first contact via a meeting request workflow with NDA acceptance, and schedule external meetings.

The platform does not provide financial transactions, contract management, medical advice, file-sharing or document repository functionality, project management features, or storage of patient data or confidential intellectual property.

1.3 Definitions and Abbreviations

Term	Definition
SRS	Software Requirements Specification
SDD	Software Design Document
RBAC	Role-Based Access Control
GDPR	General Data Protection Regulation
NDA	Non-Disclosure Agreement
WCAG	Web Content Accessibility Guidelines
JWT	JSON Web Token
API	Application Programming Interface
REST	Representational State Transfer
CRUD	Create, Read, Update, Delete
HTTPS	Hypertext Transfer Protocol Secure
Post	A structured announcement created by a user seeking collaboration partners
Meeting Request	A formal request sent by an interested user to the post owner to schedule an external meeting
Partner Found	A post status indicating that the post owner has found a suitable collaboration partner

1.4 Intended Audience

This document is intended for the SENG 384 course instructor and teaching assistants (for evaluation purposes), development team members (for implementation guidance), quality assurance

and testing personnel (for verification and validation), and project stakeholders who need to understand the system requirements.

2. Overall Description

2.1 Product Perspective

Multidisciplinary innovation in healthcare technology requires rapid and structured access to complementary expertise. Engineers developing healthcare technologies need clinical domain knowledge, workflow understanding, validation processes, and ethical approval pathways to transform concepts into viable solutions. Conversely, healthcare professionals frequently generate compelling innovation ideas but lack the engineering competence and technical infrastructure to implement them effectively.

The current landscape forces innovators to depend largely on personal networks — an inefficient model that slows innovation and limits potential partnerships to those with existing connections. The HEALTH AI platform eliminates this randomness in interdisciplinary collaboration by providing structured announcement-based partner discovery, secure first-contact initiation, controlled disclosure of ideas, a transparent meeting workflow, and clear closure of partner requests.

HEALTH AI is fundamentally a user experience platform. The system must feel safe, professional, simple, structured, and trustworthy. Users must understand the platform's purpose within 20 seconds and be able to post an announcement without confusion.

2.2 User Roles

Role	Description	Key Permissions
Engineer	A technical professional (e.g., software engineer, biomedical engineer) who registers on the platform to find healthcare domain expertise for their health-tech projects.	Create/edit/close posts; browse healthcare professional posts; send meeting requests; accept/decline meeting requests on own posts; manage own profile; express interest in posts
Healthcare Professional	A medical professional (e.g., doctor, nurse, clinical researcher) who registers on the platform to find engineering expertise or to offer clinical domain knowledge.	Create/edit/close posts; browse engineer posts; send meeting requests; accept/decline meeting requests on own posts; manage own profile; express interest in posts
Admin	A platform administrator responsible for overseeing platform operations, user management, and content moderation.	All standard user permissions; view/manage all users; suspend/deactivate accounts; view/filter/remove any post; access activity logs and statistics; export logs as CSV

2.3 Assumptions and Dependencies

It is assumed that all users possess a valid institutional .edu email address (e.g., .edu, .edu.tr). It is assumed that users have access to a modern web browser (Chrome, Firefox, Edge, Safari – latest versions) and a stable internet connection. External meeting platforms (Zoom, Microsoft Teams, Google Meet, etc.) are independently managed by users; the platform only facilitates scheduling. An external SMTP email service is required for sending verification emails and notifications. Users are expected to complete their profile after a quick initial registration to enable effective matching. The platform relies on honest self-declaration of roles; however, .edu email verification provides a basic trust layer. Fake .edu email addresses are a known risk; the system should implement additional verification measures (e.g., domain whitelist, manual admin approval for suspicious registrations).

2.4 Constraints

No patient data shall be stored on the platform at any time. No file uploads or document repository functionality shall be provided. No financial transactions or contract management shall be supported. No medical advice shall be provided through the platform. Registration is limited to institutional .edu email addresses only. Meetings are conducted externally; the platform does not host or record meetings. The platform facilitates first contact only; ongoing project collaboration tools are out of scope. NDA/confidentiality acceptance is required before initiating a meeting request conversation. Engineers shall not be able to view a directory or list of healthcare professionals (and vice versa); discovery happens only through posts.

3. Functional Requirements

3.1 User Registration & Access Control

ID	Requirement Description	Priority	Source
FR-01	The system shall only allow registration with institutional .edu email addresses. Personal email providers (Gmail, Outlook, Yahoo, etc.) shall be rejected.	High	Brief 4.1
FR-02	The system shall implement an email verification mechanism. A verification link shall be sent to the user's registered email, and the account shall not be activated until the link is confirmed.	High	Brief 4.1
FR-03	The user shall select a role during registration: Engineer or Healthcare Professional. The Admin role shall be assignable only by existing admins.	High	Brief 4.1
FR-04	The system shall enforce Role-Based Access Control (RBAC) such that users can only access functionalities permitted for their assigned role.	High	Brief 4.1
FR-05	The system shall provide a quick registration flow (email, password, role selection) followed by a separate profile completion step (name, institution, expertise areas, city, short bio).	Medium	Brief 4.1, Notes
FR-06	The system shall allow users to create and manage a profile card containing their name, institution, expertise areas, city, and a short bio. This profile card shall be visible to other users when they view the user's posts.	Medium	Notes
FR-07	The system shall implement measures to mitigate fake .edu email registration (e.g., domain whitelist, suspicious registration flagging for admin review).	Medium	Notes
FR-08	The system shall allow users to log out, terminating their active session.	High	Brief 4.1
FR-09	Engineers shall not be able to view a list/directory of healthcare professionals. Discovery shall occur only through posts.	High	Notes

3.2 Post Management

ID	Requirement Description	Priority	Source
FR-10	The system shall allow authenticated users to create a new post containing: Title, Working Domain, Short Explanation, Required Expertise (medical or engineering), Project Stage, Level of Commitment, Estimated Collaboration Type (advisor / co-founder / research partner), Confidentiality Level (public short pitch / details discussed in meeting only), Expiry Date, Country, and City.	High	Brief 4.2
FR-11	The system shall allow post owners to edit their posts while the post is in Draft or Active status.	High	Brief 4.2
FR-12	The system shall allow post owners to change the post status to "Partner Found", which closes the post and prevents new meeting requests.	High	Brief 4.2
FR-13	The system shall support the following post lifecycle states: Draft, Active, Meeting Scheduled, Partner Found (Closed), and Expired.	High	Brief 4.2

FR-14	The system shall allow users to save a post as Draft before publishing. Draft posts shall not be visible to other users.	Medium	Brief 4.2
FR-15	The system shall automatically transition a post to “Expired” status when its expiry date is reached.	Medium	Brief 4.2
FR-16	The system shall provide an auto-close option that automatically closes a post after a specified condition is met (e.g., expiry date).	Low	Brief 4.2
FR-17	The system shall NOT allow users to upload technical documentation, patient data, or any files to the platform.	High	Brief 4.2
FR-18	Post creation shall be achievable in 3 steps or fewer (create → fill fields → publish).	High	Brief 5.4
FR-19	For engineer posts, the form shall include a field for “High-level idea (without revealing sensitive details)”. For healthcare professional posts, the form shall include a field for “Desired technical expertise”.	Medium	Brief 3.1, 3.2
FR-19a	The system shall display the most recent posts at the top of the listing (newest first).	Medium	Notes

3.3 Search & Matching

ID	Requirement Description	Priority	Source
FR-20	The system shall allow users to filter posts by: Domain, Required Expertise, City, Country, Project Stage, and Status.	High	Brief 4.3
FR-21	The system shall highlight city-based matches to help users discover potential partners in the same geographic area.	Medium	Brief 4.3
FR-22	The system shall provide a match explanation for relevant results (e.g., shared expertise tags, same city).	Low	Brief 4.3
FR-23	Search results shall be returned within 1.5 seconds.	High	Brief 5.3
FR-24	Healthcare professionals shall be able to browse and filter engineer posts. Engineers shall be able to browse and filter healthcare professional posts.	High	Brief 3.1, 3.2
FR-25	During registration or profile completion, healthcare professionals shall be able to specify their interest areas; the system may use these to suggest relevant posts.	Low	Notes

3.4 Meeting Request Workflow

ID	Requirement Description	Priority	Source
FR-30	The system shall allow an interested user to send a meeting request to a post owner. The meeting request shall include a short message explaining the user’s interest and relevant experience.	High	Brief 4.4
FR-31	Before sending a meeting request, the system shall require the requesting user to accept a confidentiality/NDA agreement (e.g., “Conversations and shared information are your responsibility”).	High	Brief 4.4, Notes
FR-32	The post owner shall be able to accept or decline a meeting request.	High	Brief 4.4

FR-33	Upon acceptance, the post owner shall be able to propose meeting time slots (date and time). The requesting user shall be able to select from the proposed time slots.	High	Brief 4.4
FR-34	The system shall allow either party to cancel a meeting request before the meeting occurs.	Medium	Brief 4.4
FR-35	Meetings shall occur externally (Zoom, Teams, etc.). The platform shall not host, record, or store any meeting content.	High	Brief 4.4
FR-36	Sending a meeting request shall be achievable in under 30 seconds.	High	Brief 5.4
FR-37	The system shall provide a chat/message box within the meeting request context for users to coordinate external meeting details (e.g., sharing a Zoom link).	Medium	Notes
FR-38	The system shall send email notifications to the post owner when a new meeting request is received, and to the requester when their request is accepted/declined.	High	Notes

3.5 Administrative Dashboard

ID	Requirement Description	Priority	Source
FR-40	The admin dashboard shall be accessible only to users with the Admin role.	High	Brief 4.5
FR-41	Admins shall be able to view all posts on the platform with filtering by city, domain, and status.	High	Brief 4.5.1
FR-42	Admins shall be able to remove inappropriate posts from the platform.	High	Brief 4.5.1
FR-43	Admins shall be able to view the lifecycle history of any post (status transitions with timestamps).	Medium	Brief 4.5.1
FR-44	Admins shall be able to view platform-wide statistics (e.g., total posts, active posts, meetings scheduled, partners found).	Medium	Brief 4.5.1
FR-45	Admins shall be able to view all registered users with filtering by role.	High	Brief 4.5.2
FR-46	Admins shall be able to view profile completeness for each user.	Low	Brief 4.5.2
FR-47	Admins shall be able to suspend or deactivate user accounts.	High	Brief 4.5.2
FR-48	Admins shall be able to view activity metrics per user (e.g., number of posts, meeting requests sent/received).	Medium	Brief 4.5.2

3.6 Activity Logging & Audit Trail

ID	Requirement Description	Priority	Source
FR-50	The system shall log the following events: login, logout, failed login attempts, post creation, post editing, post closure, meeting request sent/accepted/declined/cancelled, admin actions (post removal, user suspension), and security events.	High	Brief 4.6.1
FR-51	Each log entry shall contain: Timestamp, User ID, Role, Action Type, Target Entity, and Result Status. IP address logging shall be optional and GDPR-compliant.	High	Brief 4.6.2

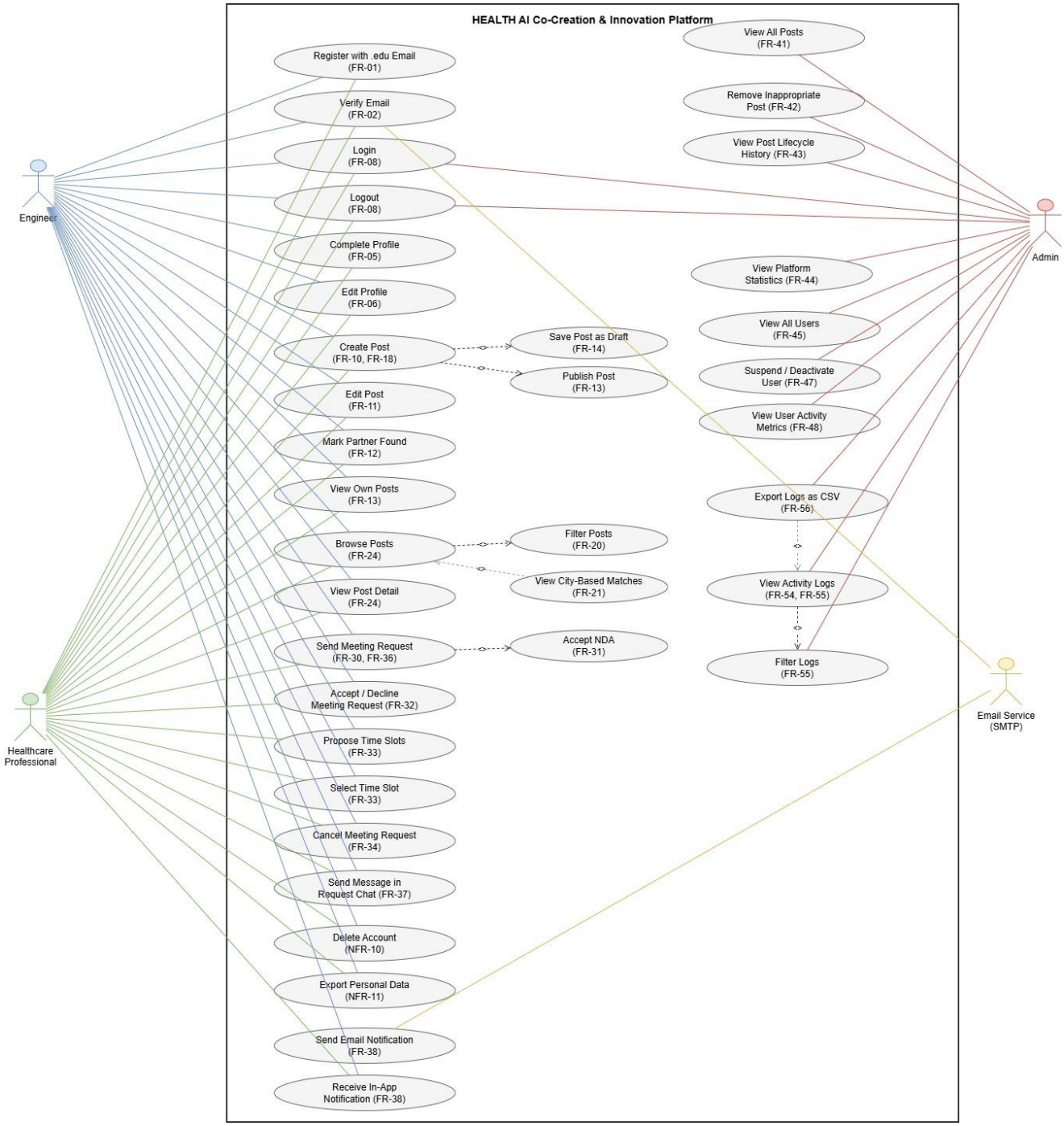
FR-52	Logs shall not contain any patient data. Data minimization principles shall be applied.	High	Brief 4.6.3
FR-53	Logs shall be tamper-resistant and stored with a defined retention period of 24 months.	Medium	Brief 4.6.3
FR-54	Log access shall be restricted to Admin users only.	High	Brief 4.6.3
FR-55	The admin log dashboard shall allow filtering logs by user, date range, and action type.	High	Brief 4.6.4
FR-56	Admins shall be able to export filtered logs in CSV format.	Medium	Brief 4.6.4
FR-57	The system should support anomaly pattern detection in logs (e.g., excessive failed login attempts, unusual activity spikes).	Low	Brief 4.6.4

4. Non-Functional Requirements

ID	Requirement	Metric / Target	Category
NFR-01	Search results shall be returned within a specified time.	< 1.5 seconds	Performance
NFR-02	Page load time shall not exceed a specified threshold.	< 3 seconds	Performance
NFR-03	The system shall support a specified number of concurrent users without performance degradation.	1000 concurrent users	Performance
NFR-04	All communication between client and server shall use HTTPS.	HTTPS enforced on all endpoints	Security
NFR-05	User passwords shall be stored using a strong, salted hashing algorithm.	bcrypt or argon2	Security
NFR-06	The system shall implement secure session management with automatic session timeout after inactivity.	Timeout after 30 min	Security
NFR-07	The system shall implement rate limiting on authentication endpoints to prevent brute-force attacks.	Max 10 failed login attempts per 15 min per IP	Security
NFR-08	The system shall implement anti-bot protection on registration and login forms.	CAPTCHA or equivalent	Security
NFR-09	The system shall collect only the minimum data necessary for its functionality (data minimization).	GDPR Art. 5(1)(c)	GDPR / Privacy
NFR-10	Users shall have the right to delete their account and all associated personal data permanently.	Full deletion within 72h	GDPR / Privacy
NFR-11	Users shall be able to export their personal data in a portable, machine-readable format.	JSON or CSV export	GDPR / Privacy
NFR-12	The platform shall display a clear privacy policy explaining data collection, usage, and retention.	Accessible from registration and footer	GDPR / Privacy
NFR-13	No patient data shall be stored on the platform at any point.	Zero patient data records	GDPR / Privacy
NFR-14	A new user shall be able to understand the platform's purpose within 20 seconds of landing on the homepage.	Usability testing target	Usability
NFR-15	Post creation shall be completable in 3 steps or fewer.	≤ 3 steps	Usability
NFR-16	Posting an announcement shall take less than 3 minutes from start to publish.	< 3 minutes	Usability
NFR-17	Sending a meeting request shall take less than 30 seconds.	< 30 seconds	Usability
NFR-18	The platform shall use plain, jargon-free language. Tooltips shall be provided for terms like IP, NDA.	All UI text reviewed	Usability

NFR-19	Clear, color-coded status indicators shall be used for post lifecycle states.	Distinct colors per status	Usability
NFR-20	The platform shall comply with WCAG 2.1 Level AA accessibility guidelines.	WCAG 2.1 AA	Accessibility
NFR-21	The platform shall be fully responsive and functional on mobile devices, tablets, and desktops.	Min viewport: 320px	Accessibility

5. Use Cases


HEALTH AI Co-Creation & Innovation Platform
Engineer
Healthcare Professional
Admin
Email Service (SMTP)
Register with .edu Email (FR-01)
Verify Email (FR-02)
Login (FR-08)
Logout (FR-08)
Complete Profile (FR-05)
Edit Profile (FR-06)
Create Post (FR-10, FR-18)
Save Post as Draft (FR-14)
Publish Post (FR-13)
Edit Post (FR-11)
Mark Partner Found (FR-12)
View Own Posts (FR-13)
Browse Posts (FR-24)
Filter Posts (FR-20)
View City-Based Matches (FR-21)
View Post Detail (FR-24)
Send Meeting Request (FR-30, FR-36)
Accept NDA (FR-31)
Accept / Decline Meeting Request (FR-32)
Propose Time Slots (FR-33)
Select Time Slot (FR-33)
Cancel Meeting Request (FR-34)
Send Message in Request Chat (FR-37)
Delete Account (NFR-10)
Export Personal Data (NFR-11)
Send Email Notification (FR-38)
Receive In-App Notification (FR-38)
View All Posts (FR-41)
Remove Inappropriate Post (FR-42)
View Post Lifecycle History (FR-43)
View Platform Statistics (FR-44)
View All Users (FR-45)
Suspend / Deactivate User (FR-47)
View User Activity Metrics (FR-48)
Export Logs as CSV (FR-56)
View Activity Logs (FR-54, FR-55)
Filter Logs (FR-55)

5.1 UC-01: Engineer Creates a Post

Field	Detail
Name	UC-01: Engineer Creates a Post
Actor(s)	Engineer
Precondition	The engineer must be logged in and have a verified account.
Main Flow	1. The engineer clicks the “+” or “New Post” button on the Dashboard. 2. The system displays the post creation form with fields: Title, Working Domain, Short Explanation, Required Healthcare Expertise, Project Stage, Level of Commitment, High-level Idea, Estimated Collaboration Type, Confidentiality Level, Expiry Date, Country, City. 3. The engineer fills in all required fields. 4. The engineer selects “Publish” or “Save as Draft”. 5. If “Publish” is selected, the system validates the form, saves the post with “Active” status, and displays it on the public feed.
Postcondition	The post is saved in the system. If published, it is visible to healthcare professionals on the platform.
Alternative Flow	3a. If a required field is left blank, the system highlights the missing field and displays an error message. 4a. If “Save as Draft” is selected, the post is saved with “Draft” status and is only visible to the owner.

5.2 UC-02: Healthcare Professional Sends Meeting Request

Field	Detail
Name	UC-02: Healthcare Professional Sends Meeting Request
Actor(s)	Healthcare Professional
Precondition	The healthcare professional must be logged in. The target post must be in “Active” status.
Main Flow	1. The healthcare professional browses or filters the post feed. 2. Clicks on a post to view details. 3. Clicks “Express Interest” / “Request Meeting”. 4. The system presents the NDA/Confidentiality acceptance screen. 5. Reads and accepts the NDA terms. 6. Writes a short message explaining interest and relevant experience. 7. Clicks “Send Request”. 8. The system saves the meeting request with “Pending” status and sends an email notification to the post owner.
Postcondition	A meeting request is created and the post owner is notified via email and in-app notification.
Alternative Flow	5a. If the NDA is declined, the process is cancelled. 7a. If the message field is empty, the system prompts the user to provide a message.

5.3 UC-03: Admin Removes Inappropriate Post

Field	Detail
Name	UC-03: Admin Removes Inappropriate Post
Actor(s)	Admin
Precondition	The admin must be logged in with Admin role. The target post exists on the platform.
Main Flow	1. The admin navigates to Admin Panel → Posts. 2. Browses or filters posts (by status, domain, date). 3. Clicks on a post to view details and lifecycle history. 4. Determines the post violates platform guidelines. 5. Clicks “Remove Post”. 6. The system prompts confirmation and optionally a reason. 7. The admin confirms. 8. The system marks the post as removed, logs the action, and notifies the post owner.
Postcondition	The post is removed from public view. An activity log entry is created. The post owner is notified.
Alternative Flow	7a. If the admin cancels confirmation, the post remains unchanged.

5.4 UC-04: User Marks Partner Found

Field	Detail
Name	UC-04: User Marks Partner Found
Actor(s)	Engineer or Healthcare Professional (post owner)
Precondition	The user must be logged in and own the post. The post must be in “Active” or “Meeting Scheduled” status.
Main Flow	1. The user navigates to their post (from Dashboard or “My Posts”). 2. Clicks the “Mark as Partner Found” button. 3. The system displays a confirmation dialog. 4. The user confirms. 5. The system updates the post status to “Partner Found (Closed)”. 6. The post no longer accepts new meeting requests. 7. All pending meeting requests are notified that the post has been closed.
Postcondition	The post status is “Partner Found (Closed)”. No new meeting requests can be submitted.
Alternative Flow	4a. If the user cancels confirmation, the post status remains unchanged.

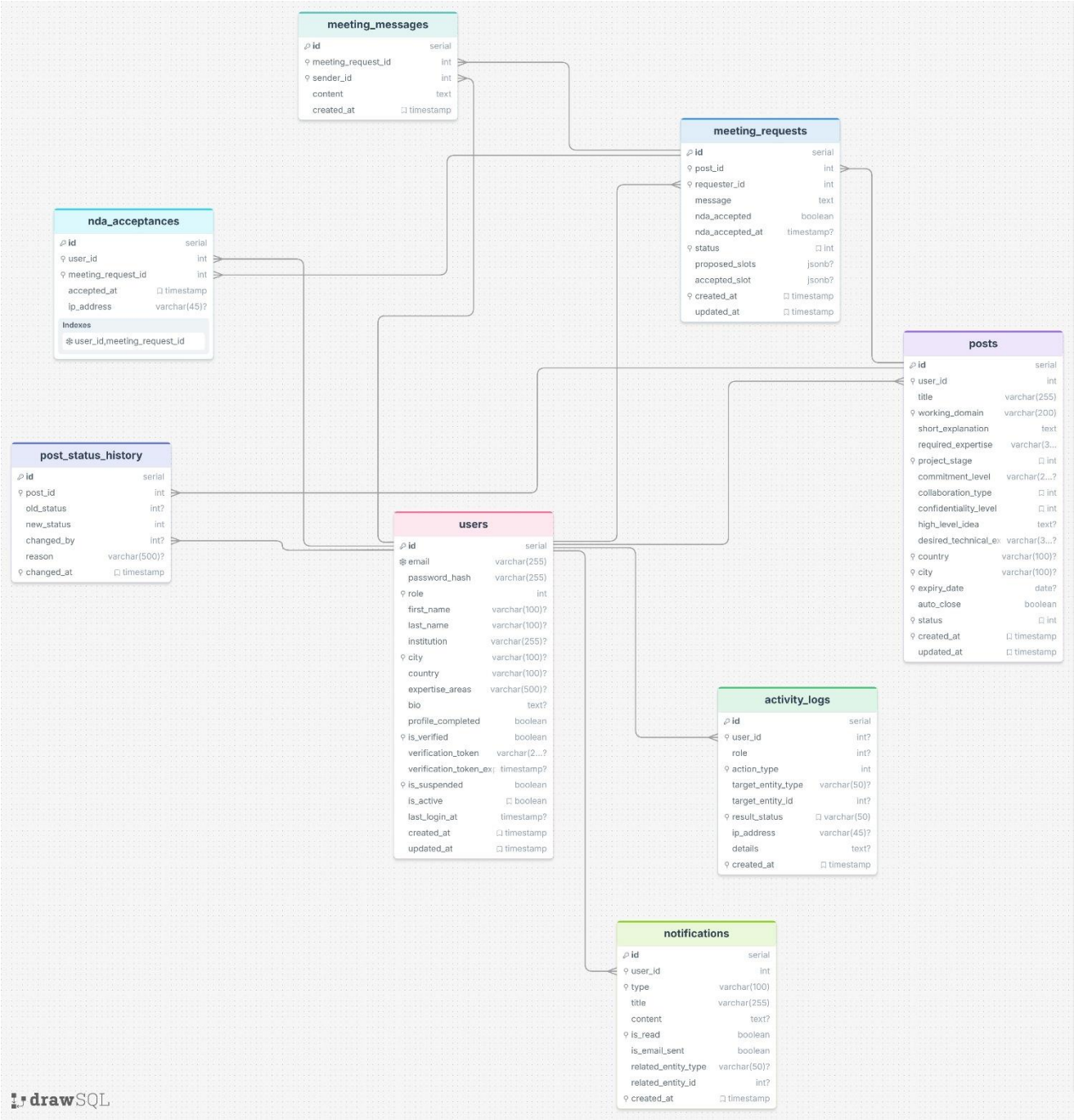
5.5 UC-05: Healthcare Professional Creates a Post

Field	Detail
Name	UC-05: Healthcare Professional Creates a Post
Actor(s)	Healthcare Professional
Precondition	The healthcare professional must be logged in and have a verified account.
Main Flow	1. Clicks the “+” or “New Post” button on the Dashboard. 2. The system displays the post creation form with fields: Title, Medical Field, Short Explanation, Desired Technical Expertise, Project Stage, Level of Commitment, Estimated Collaboration Type, Confidentiality Level, Expiry Date, Country, City. 3. Fills in all required fields. 4. Selects “Publish” or “Save as Draft”. 5. If “Publish” is selected, the system validates and saves with “Active” status.
Postcondition	The post is saved. If published, it is visible to engineers on the platform.
Alternative Flow	3a. If a required field is left blank, the system highlights the missing field. 4a. If “Save as Draft” is selected, the post is saved with “Draft” status.

5.6 UC-06: User Registers to the Platform

Field	Detail
Name	UC-06: User Registers to the Platform
Actor(s)	Engineer or Healthcare Professional
Precondition	The user has a valid institutional .edu email address.
Main Flow	1. Navigates to the platform homepage. 2. Clicks “Register”. 3. Enters .edu email address and creates a password. 4. Selects role (Engineer or Healthcare Professional). 5. Clicks “Register”. 6. The system validates the email domain. 7. Sends a verification email. 8. The user clicks the verification link. 9. Account is activated. Redirected to profile completion. 10. Fills in profile details (name, institution, expertise areas, city, short bio).
Postcondition	The user account is created, verified, and the profile is completed.
Alternative Flow	6a. If the email is not .edu, registration is rejected. 8a. If the verification link expired, the user can request a new one. 10a. Profile completion can be skipped but matching effectiveness is reduced.

6. Data Model


meeting_messages
id serial
meeting_request_id int
sender_id int
content text
created_at timestamp
meeting_requests
id serial
post_id int
requester_id int
message text
nda_accepted boolean
nda_accepted_at timestamp?
status int
proposed_slots jsonb?
accepted_slot jsonb?
created_at timestamp
updated_at timestamp
nda_acceptances
id serial
user_id int
meeting_request_id int
accepted_at timestamp
ip_address varchar(45)?
Indexes
user_id,meeting_request_id
posts
id serial
user_id int
title varchar(255)
working_domain varchar(200)
short_explanation text
required_expertise varchar(3...
project_stage int
commitment_level varchar(2...?
collaboration_type int
confidentiality_level int
high_level_idea text?
desired_technical_ex varchar(3...?
country varchar(100)?
city varchar(100)?
expiry_date date?
auto_close boolean
status int
created_at timestamp
updated_at timestamp
post_status_history
id serial
post_id int
old_status int?
new_status int
changed_by int?
reason varchar(500)?
changed_at timestamp
users
id serial
email varchar(255)
password_hash varchar(255)
role int
first_name varchar(100)?
last_name varchar(100)?
institution varchar(255)?
city varchar(100)?
country varchar(100)?
expertise_areas varchar(500)?
bio text?
profile_completed boolean
is_verified boolean
verification_token varchar(2...?
verification_token_ex timestamp?
is_suspended boolean
is_active boolean
last_login_at timestamp?
created_at timestamp
updated_at timestamp
activity_logs
id serial
user_id int?
role int?
action_type int
target_entity_type varchar(50)?
target_entity_id int?
result_status varchar(50)
ip_address varchar(45)?
details text?
created_at timestamp
notifications
id serial
user_id int
type varchar(100)
title varchar(255)
content text?
is_read boolean
is_email_sent boolean
related_entity_type varchar(50)?
related_entity_id int?
created_at timestamp
drawSQL

Entity	Key Fields	Relationships
User	id (PK), email (UNIQUE, .edu check), password_hash, role (engineer / healthcare / admin), first_name, last_name, institution, city, country, expertise_areas, bio, profile_completed, is_verified, is_suspended, is_active, created_at, updated_at	One User → Many Posts; One User → Many MeetingRequests (as requester); One User → Many ActivityLogs; One User → Many Notifications
Post	id (PK), user_id (FK → User), title, working_domain, short_explanation, required_expertise, project_stage, commitment_level, collaboration_type, confidentiality_level, high_level_idea, desired_technical_expertise, expiry_date, auto_close, country, city, status (draft / active / meeting_scheduled / partner_found / expired), created_at, updated_at	One Post → One User (owner); One Post → Many MeetingRequests; One Post → Many PostStatusHistory
PostStatusHistory	id (PK), post_id (FK → Post), old_status, new_status, changed_by (FK → User), reason, changed_at	One PostStatusHistory → One Post; One PostStatusHistory → One User
MeetingRequest	id (PK), post_id (FK → Post), requester_id (FK → User), message, status (pending / accepted / declined / scheduled / completed / cancelled), nda_accepted, nda_accepted_at, proposed_slots (JSON), accepted_slot, created_at, updated_at	One MeetingRequest → One Post; One MeetingRequest → One User (requester); One MeetingRequest → Many MeetingMessages
MeetingMessage	id (PK), meeting_request_id (FK → MeetingRequest), sender_id (FK → User), content, created_at	One MeetingMessage → One MeetingRequest; One MeetingMessage → One User (sender)
NDAAcceptance	id (PK), user_id (FK → User), meeting_request_id (FK → MeetingRequest), accepted_at, ip_address (optional)	One NDAAcceptance → One User; One NDAAcceptance → One MeetingRequest
Notification	id (PK), user_id (FK → User), type, title, content, is_read, is_email_sent, related_entity_type, related_entity_id, created_at	One Notification → One User
ActivityLog	id (PK), user_id (FK → User), role, action_type, target_entity_type, target_entity_id, result_status, ip_address (optional, GDPR-compliant), details, created_at	One ActivityLog → One User

7. Interface Requirements

7.1 User Interface (UI)

The platform UI shall be clean, minimal, and focused on task completion with minimum clicks. The following screens are required:

Screen	Description	Access
Landing Page	Platform introduction explaining the purpose clearly within 20 seconds. Contains “Register” and “Login” buttons.	Public
Registration Page	Email input (.edu validation), password creation, role selection. Minimal fields for quick registration.	Public
Login Page	Email and password input. Link to “Forgot Password” and “Resend Verification Email”.	Public
Dashboard	Main screen after login. Shows a feed of relevant posts (newest first), a prominent “+” or “New Post” button, filter/search bar, and notification indicator.	Authenticated
Post Creation Form	A step-based form (≤ 3 steps) with all required post fields. Role-specific fields shown based on user type.	Authenticated
Post Detail Page	Full post information, post owner’s profile card, status badge, and “Express Interest” / “Request Meeting” button.	Authenticated
My Posts	Lists all posts created by the logged-in user with status badges. Allows editing and status changes.	Authenticated
Meeting Request View	Meeting request details, requester’s profile card, message, NDA status, proposed time slots, and message box.	Authenticated
Profile Page	View and edit profile information. Account deletion and data export options.	Authenticated
Admin Panel – Posts	Table/list of all posts with filters (city, domain, status, date). Actions: view details, remove post.	Admin
Admin Panel – Users	Table/list of all users with filters (role, status). Actions: view profile, suspend/deactivate.	Admin
Admin Panel – Logs	Searchable and filterable activity log table. CSV export button.	Admin
Admin Panel – Statistics	Platform-wide metrics: total users, total posts, active posts, meetings scheduled, partners found.	Admin

[VISUAL: En az 3-4 ana ekran için wireframe / mockup veya ekran görüntüsü buraya eklenecek (Registration, Dashboard, Post Creation, Post Detail + Meeting Request).]

Key UI Principles:

Simple, minimal screens with minimum clicks to achieve goals. No cluttered layouts. The “+” / “New Post” button must be immediately visible on the dashboard upon login. Newest announcements displayed at the top. Clear navigation flow. No unnecessary images or decorative elements. Color-coded status badges: Gray = Draft, Green = Active, Blue = Meeting Scheduled, Purple = Partner Found, Red = Expired.

7.2 External System Interfaces

External System	Purpose	Integration Method
SMTP Email Service (e.g., SendGrid, AWS SES, Mailgun)	Sending verification emails, notification emails (meeting requests, status changes, account activity)	SMTP API / REST API
External Meeting Platforms (Zoom, Teams, Google Meet)	Users share meeting links manually via the in-platform message box. No direct integration.	Link sharing only (no API integration)

8. Requirements Traceability Matrix

Req. ID	Requirement Summary	Source (Brief)	Related Use Case
FR-01	.edu email restriction	Brief 4.1	UC-06
FR-02	Email verification mechanism	Brief 4.1	UC-06
FR-03	Role selection during registration	Brief 4.1	UC-06
FR-04	Role-Based Access Control (RBAC)	Brief 4.1	All UCs
FR-05	Quick registration + profile completion	Brief 4.1, Notes	UC-06
FR-06	User profile card	Notes	UC-01, UC-02, UC-05
FR-07	Fake .edu email mitigation	Notes	UC-06
FR-08	Logout functionality	Brief 4.1	All UCs
FR-09	No user directory browsing	Notes	UC-02
FR-10	Post creation with all fields	Brief 4.2	UC-01, UC-05
FR-11	Post editing (Draft/Active)	Brief 4.2	UC-01, UC-05
FR-12	Mark post as Partner Found	Brief 4.2	UC-04
FR-13	Post lifecycle states	Brief 4.2	UC-01, UC-04, UC-05
FR-14	Save as Draft	Brief 4.2	UC-01, UC-05
FR-15	Auto-expiry on expiry date	Brief 4.2	—
FR-16	Auto-close option	Brief 4.2	—
FR-17	No file uploads	Brief 4.2	All UCs
FR-18	Post creation ≤ 3 steps	Brief 5.4	UC-01, UC-05
FR-19	Role-specific post form fields	Brief 3.1, 3.2	UC-01, UC-05
FR-19a	Newest posts first	Notes	UC-02
FR-20	Search/filter by domain, city, etc.	Brief 4.3	UC-02
FR-21	City-based match highlighting	Brief 4.3	UC-02
FR-22	Match explanation	Brief 4.3	UC-02
FR-23	Search < 1.5 sec	Brief 5.3	UC-02
FR-24	Cross-role post browsing	Brief 3.1, 3.2	UC-02
FR-25	Interest area-based suggestions	Notes	UC-02
FR-30	Meeting request with message	Brief 4.4	UC-02
FR-31	NDA acceptance before request	Brief 4.4, Notes	UC-02
FR-32	Accept/decline meeting request	Brief 4.4	UC-02
FR-33	Time slot proposal and selection	Brief 4.4	UC-02
FR-34	Cancel meeting request	Brief 4.4	UC-02
FR-35	External meetings only	Brief 4.4	UC-02
FR-36	Meeting request < 30 sec	Brief 5.4	UC-02
FR-37	Message box for coordination	Notes	UC-02
FR-38	Email notifications for requests	Notes	UC-02

FR-40	Admin dashboard access control	Brief 4.5	UC-03
FR-41	Admin view/filter all posts	Brief 4.5.1	UC-03
FR-42	Admin remove post	Brief 4.5.1	UC-03
FR-43	Post lifecycle history	Brief 4.5.1	UC-03
FR-44	Platform statistics	Brief 4.5.1	—
FR-45	Admin view/filter all users	Brief 4.5.2	—
FR-46	Profile completeness view	Brief 4.5.2	—
FR-47	Suspend/deactivate accounts	Brief 4.5.2	—
FR-48	User activity metrics	Brief 4.5.2	—
FR-50	Comprehensive event logging	Brief 4.6.1	All UCs
FR-51	Log data fields	Brief 4.6.2	All UCs
FR-52	No patient data in logs	Brief 4.6.3	All UCs
FR-53	Tamper-resistant logs, 24-month retention	Brief 4.6.3	—
FR-54	Admin-only log access	Brief 4.6.3	UC-03
FR-55	Filter logs by user/date/action	Brief 4.6.4	UC-03
FR-56	Export logs as CSV	Brief 4.6.4	UC-03
FR-57	Anomaly pattern detection	Brief 4.6.4	—

9. Appendices

[Appendix A: Use Case Diagram]

[Appendix B: Wireframes / Mockups]

[Appendix C: ER Diagram]